

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2179

Fuller 138 kV
March 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2179. The Interconnection Customer proposes a 136.9 MW solar photovoltaic facility interconnecting at the Fuller 138 kV substation in the Kentucky area. The request was received on September 09, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 588 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,721,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Twisp / Vly Springs 138 KV line
Contingency: Ss3144 Tap / Greenwater 138 KV Line
Loading: 124.4% (Rating: 192 MVA)
DFAX: 100.0%
Pre-project loading: 53.2%
2. Facility: Tracy Defense Depot / Alder Creek-Bradshaw 138 KV
Contingency: Belson / Meadow Brook 138 KV
Loading: 109.0% (Rating: 233 MVA)
DFAX: 48.0%
Pre-project loading: 80.9%
3. Facility: Alder Creek-Bradshaw / La Fresa 138 KV line
Contingency: Belson / Meadow Brook 138 KV
Loading: 106.1% (Rating: 225 MVA)
DFAX: 48.0%
Pre-project loading: 76.9%
4. Facility: La Fresa / Munk 138 KV line
Contingency: Belson / Meadow Brook Substation 138 KV
Loading: 103.5% (Rating: 215 MVA)
DFAX: 48.0%
Pre-project loading: 73.0%
5. Facility: Keifer-Jackson / Beaver Dam E 138 KV
Contingency: Tyrana Substation / Lg&E; Canal 138 KV
Loading: 102.7% (Rating: 251 MVA)
DFAX: 6.9%
Pre-project loading: 98.9%

6. Facility: Greenwater / Ss3144 Tap 138 KV
Contingency: Twisp Substation / Vly Springs Substation 138 KV
Loading: 102.6% (Rating: 233 MVA)
DFAX: 100.0%
Pre-project loading: 43.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Twisp / Valley Springs 138 KV	Rebuild line with higher-capacity conductor	\$6,569,000	100% (\$6,569,000)
Tracy Defense Depot / Alder Creek-Bradshaw 138 KV	Reconductor line	\$2,484,000	100% (\$2,484,000)
Alder Creek-Bradshaw / La Fresa 138 KV	Reconductor line	\$1,958,000	100% (\$1,958,000)
La Fresa / Munk 138 KV line	Reconductor line	\$1,093,000	100% (\$1,093,000)
Keifer-Jackson / Beaver Dam E 138 KV	Reconductor line	\$1,896,000	10% (\$192,000)
Greenwater / Ss3144 Tap 138 KV	Reconductor line	\$5,425,000	100% (\$5,425,000)
	Total allocated to project		\$17,721,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.